

Week 2 Assignment 2

Q1.

A bank wants to predict whether a loan application will be approved or rejected.

- Is this a classification or regression problem?
- Which algorithm among Decision Tree, KNN, and Linear Regression would be most suitable?

Q2.

Explain the difference between a feature and a label using a real-world example.

Q3.

A Decision Tree predicts that a customer will purchase a product.

Explain how the model reaches this prediction.

Q4.

Why is KNN considered a supervised learning algorithm?

Can KNN perform classification if labels are not available?

Q5.

Suppose a KNN model uses:

$K = 5$

and the nearest neighbors are:

Yes, Yes, No, Yes, No

What will be the prediction and why?

Q6.

Explain the equation of Linear Regression:

$$y = mx + b$$

m

b

What do x , y , m , and b represent?

Q7.

What is correlation?

What does each of the following values indicate?

+1

0

-1

Q8.

A disease detection model produced the following result:

Actual Condition Prediction

Disease Present Healthy

- What type of prediction is this?
- Why can this be dangerous?

Q9.

What is the purpose of a Confusion Matrix?

How is it different from a Correlation Matrix?

Q10.

Compare the following algorithms in one or two lines each:

- Decision Tree
- KNN
- SVM
- Linear Regression
- Neural Network

Focus on the main idea behind each algorithm.